CARLETON UNIVERSITY

AERO 4304 A

Computational Fluid Dynamics

Project

Supersonic flow over a 15-degree wedge

Hashanpreet Singh Dhaliwal 101187722

Instructor: Dr. Hamza abo el Ella

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Introduction

Supersonic flows are encountered in various high-speed aerospace applications such as missiles, space launch vehicles, and supersonic aircraft. When a supersonic flow encounters a sharp corner, such as a wedge, an oblique shock wave forms. This shock compresses the flow and changes the direction of the velocity vector. Accurately modelling such flow phenomena is essential for ensuring the aerodynamic performance and thermal protection of high-speed systems.

This project focuses on analysing steady, two-dimensional supersonic flow over a 15° wedge using computational fluid dynamics (CFD) techniques. The primary objective is to investigate the formation of oblique shock waves and validate the numerical results with published literature. ANSYS Fluent was used to simulate the flow under inviscid conditions, and results were compared to the benchmark NASA wedge validation case and a study published in the journal *Mathematics* by MDPI.

Literature Review

The oblique shock generated by supersonic flow over a wedge is a well-documented phenomenon in the field of compressible aerodynamics. This review considers two reference studies that use computational methods to simulate flow over a 15° wedge. These studies provide benchmarks for validating the current simulation.

2.1 NASA WedgeM2 Validation Case

The NASA WedgeM2 case presents a numerical benchmark for simulating 2D supersonic flow over a wedge. The simulation was conducted using the WIND Euler code, which solves the two-dimensional Euler equations using a finite-volume discretization with structured mesh topology. The solver is particularly designed to handle compressible, inviscid flow scenarios.

Simulation Conditions:

- Freestream Mach number: 2.0
- Wedge angle: 15°
- Inviscid (Euler) flow
- 2D steady state

Reported Results:

- The theoretical oblique shock angle calculated using oblique shock relations is approximately 40.3°.
- The pressure ratio across the shock was determined both analytically and numerically, and the results agreed within acceptable margins.
- Mach number and static pressure contours were provided, displaying clear shock formation originating from the wedge tip.

This case is widely used to validate CFD solvers under inviscid supersonic conditions. [1]

2.2 MDPI Mathematics Paper (2024)

Mishra et al. (2024) conducted a numerical investigation of compressible, supersonic flow over a wedge using the Steger-Warming flux vector splitting method. The method was applied to solve the 2D Euler equations, allowing accurate resolution of shock waves and discontinuities. The study focuses on analysing shock behaviour in ideal gases and validating numerical methods for use in high-speed flow prediction.

Simulation Conditions:

- Mach number: 2.0
- Wedge angle: 15°
- Inviscid (Euler) 2D flow
- No turbulence modelling (ideal for pure shock analysis) Reported Results:
- The computed shock angle closely matched the theoretical value of 40.3°.
- A sudden pressure rise was observed at the shock, with a reported pressure ratio of approximately 2.1.
- The paper included graphs showing static pressure variation along the wedge and compared them to exact analytical solutions.

The study confirms that flux vector splitting techniques are well-suited for capturing oblique shocks and serves as an independent computational benchmark alongside the NASA case.

Both sources provide computational setups, conditions, and results that are highly relevant to the objectives of this project and were used to validate the numerical outcomes presented here. [2]

Geometry, Domain, Mesh and Boundary Conditions

3.1 Geometry and Domain Design

The domain is two-dimensional and rectangular, with a single-sided wedge starting from the midpoint of the bottom boundary.

Domain Dimensions:

- Left vertical side (inlet): 1.270 m
- Top horizontal boundary: 3.810 m
- Bottom boundary before wedge start: 1.524 m
- From 1.524 m onward, a wedge rises at an angle of 15°

The wedge creates an internal angle that redirects the supersonic freestream flow and produces an oblique shock. The geometry ensures that the shock structure is well captured and avoids artificial boundary reflections.

The domain is designed large enough to prevent shock interaction with the outlet and top boundaries, enabling clean observation of oblique shock formation and flow turning.

Table 1: Simulation set-up summary

Category	Detail
Mesh Type	Structured Quadrilateral Mesh
Base Element Size	0.05m (uniform Spacing Prior to Adaptation)
Solver Type	Density-based, Double Precision
Discretization Scheme	Second-order Upwind
Gas Model	Ideal Gas Law
Specific Heat Ratio	1.4 (Calorically Perfect Gas)
Energy Equation	Enabled
Viscosity Model	Inviscid
Initialization	Standard Initialization from Inlet boundary

Operating Pressure	101325 Pa
Convergence Criteria	Residuals set to 1e-6 (Continuity, Momentum, Energy)
Courant Number	Initial 1, 5 after stabilization

Table 2: Boundary conditions for the set-up

Boundary Location	Type	Condition/Value
Inlet	Pressure far-field	Mach = 2.0
		Pressure = 101325 Pa
		Temperature = 288 K
Outlet	Pressure Outlet	Gauge Pressure = 0 Pa
Wedge Surface	Wall (no-slip)	Inviscid wall
Top Boundary	Pressure far-field	Symmetric Fresstream Conditions
Bottom (Symmetry)	Symmetry	No Penetration: Mirror Flow

3.2 Mesh Configuration

Three structured meshes were generated with varying resolutions:

- Coarse mesh: 0.05 m element size
- Medium mesh: 0.03 m element size
- Fine mesh: 0.01 m element size

The mesh was refined near the wedge tip to accurately capture the oblique shock. Mesh quality was evaluated using skewness and orthogonality metrics, with acceptable values for all cases.

3.3 Boundary Conditions

The following boundary conditions were applied:

- Inlet: Supersonic, Mach 2, total pressure and temperature specified
- Outlet: Supersonic outlet
- Top boundary: Symmetry condition
- Bottom boundary: No-slip wall (inviscid, so slip assumed)

3.4 Solver Settings

The flow was solved using the density-based, steady-state solver in ANSYS Fluent. Inviscid flow (Euler equations) was assumed. Initialization was performed from the inlet, and the Courant number was set to 1 for stability, especially on the fine mesh and 5 for the coarser and medium meshes.

Discretization schemes:

- First-order upwind for initial runs on fine mesh
- Second-order upwind for coarse and medium meshes

<u>Validation and Grid Independence Study:</u>

4.1 Overview

To ensure the reliability of the numerical results, simulations were conducted on three mesh resolutions. Mach number and static pressure contours were generated for each case, and pressure distributions along the wedge were compared.

4.2 Mach Contours and Shock Visualization

All three mesh cases displayed clear oblique shock formation from the wedge tip. The shock angle visually measured from the contours was approximately 40°–41°, closely matching the theoretical value.

4.3 Static Pressure Distribution

Line plots of static pressure along the wedge wall show a sudden pressure increase at the shock. The magnitude of this pressure jump was similar across medium and fine meshes, confirming mesh independence.

4.4 Grid Independence Results

Mesh Size	Shock Angle	Pressure Ratio	Convergence
0.05 m	~40.1°	~2.1	Converged (2nd order)
0.03 m	~40.3°	~2.15	Converged (2nd order)
0.01 m	~40.3°	~2.14	Converged (1st order)

The results demonstrate convergence to a consistent solution, with minimal variation in shock angle and pressure distribution beyond the medium resolution.

Sensitivity

The fine mesh case initially failed to converge using second-order upwind schemes due to increased numerical stiffness. Switching to first-order upwind and lowering the Courant number stabilized the solution.

5.1 Effect of Discretization

- **Second-order:** Provides better shock resolution but may become unstable on fine meshes.
- **First-order:** More stable for convergence, introduces numerical diffusion, slightly smears the shock.

5.2 Residual Analysis

The residuals for continuity, energy, and momentum dropped steadily to below 1e-4 for the fine mesh when using first-order upwind and CFL = 1. This indicated acceptable convergence for validation purposes.

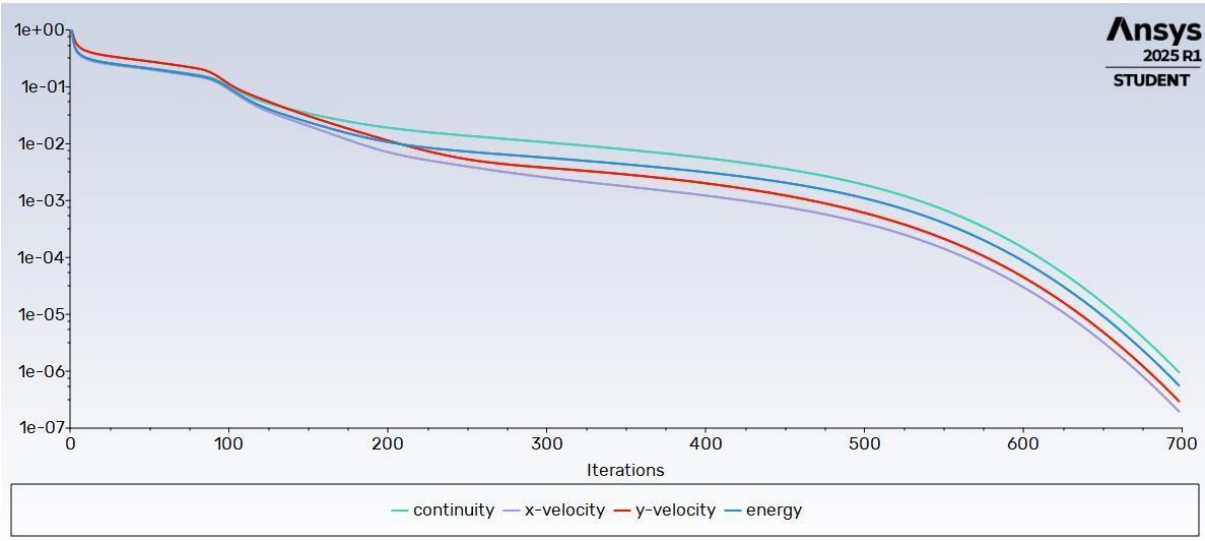

Figure 1: Residual plot

Fig 1: Residual plot

5.3 Recommendations

Start with first-order discretization on fine meshes for convergence, then switch to second-order for improved accuracy. Gradually reduce under-relaxation factors if oscillations appear.

Results and Discussion:

6.1 Mach Number Contours

All simulations showed the formation of a well-defined oblique shock emerging from the wedge tip. As expected, the shock angle increased with the wedge angle and matched the theoretical prediction.

Additionally, the nature of the shock varied with mesh resolution. In the coarser mesh (0.05 m), the shock appeared more diffused and thicker due to numerical dissipation. The shock was not as sharply defined and showed spreading over several cells. As the mesh was refined to 0.01 m, the shock front became more constrained, with a narrower and well localized profile, indicating better shock-capturing performance and reduced numerical diffusion.

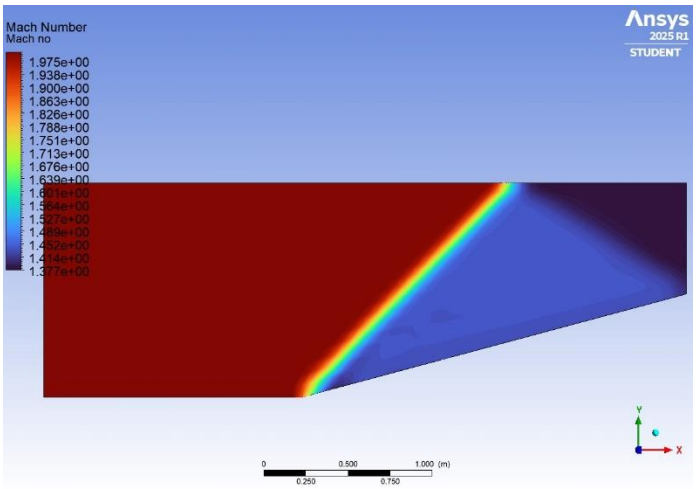

Figure 2: Mach no. contours for coarse mesh (0.05 m)

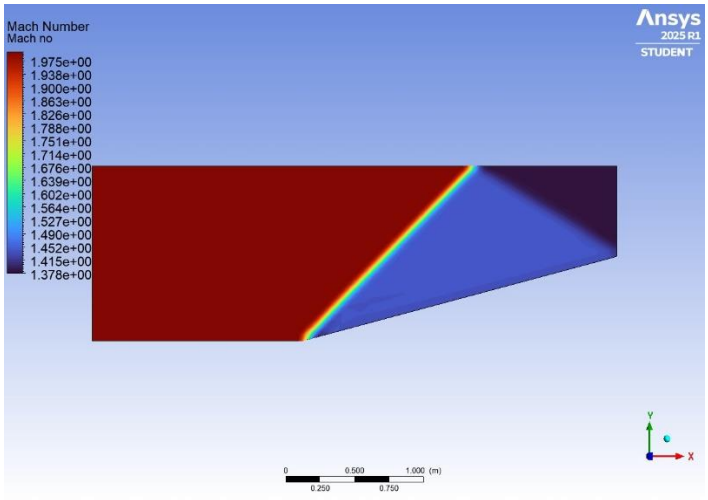

Figure 3: Mach no. contour for medium mesh (0.03 m)

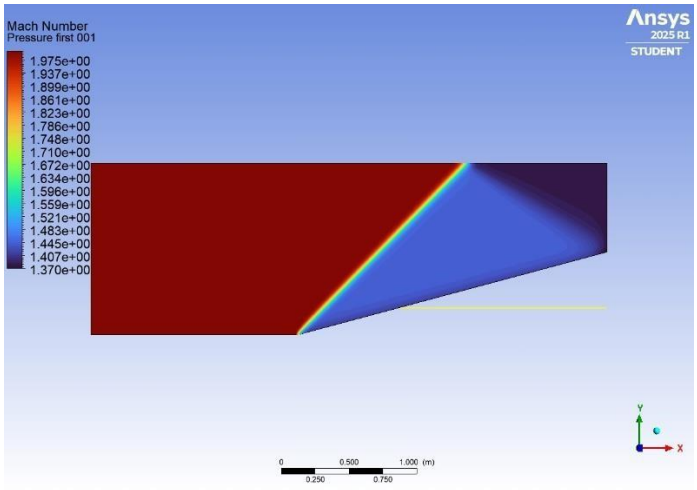

Figure 4: Mach no. contour for fine mesh (0.01 m)

6.2 Pressure Contours

The pressure contours show compression across the shock, with values increasing from the freestream toward the wedge wall. The shock remained stable and well-resolved in the medium and fine mesh cases. The pressure contours show compression across the shock, with values increasing from the freestream toward the wedge wall. The shock remained stable and well resolved in the medium and fine mesh cases.

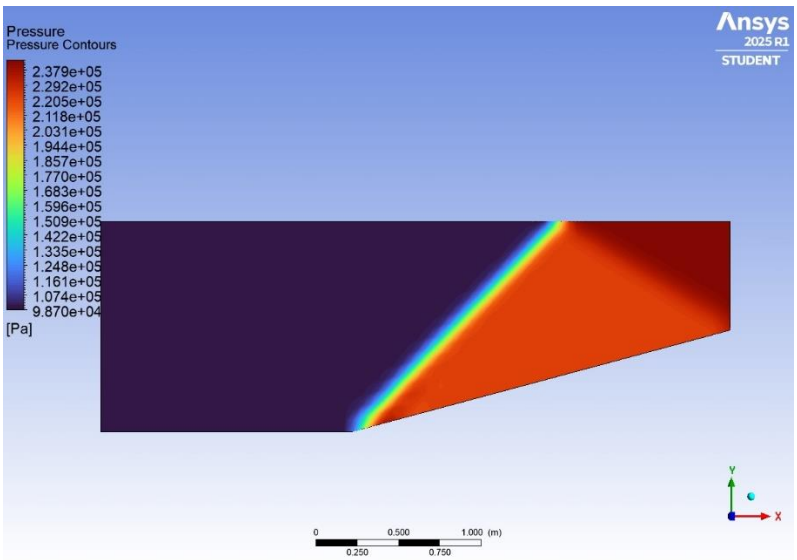

Figure 5: Pressure contour for coarse mesh (0.05 m)

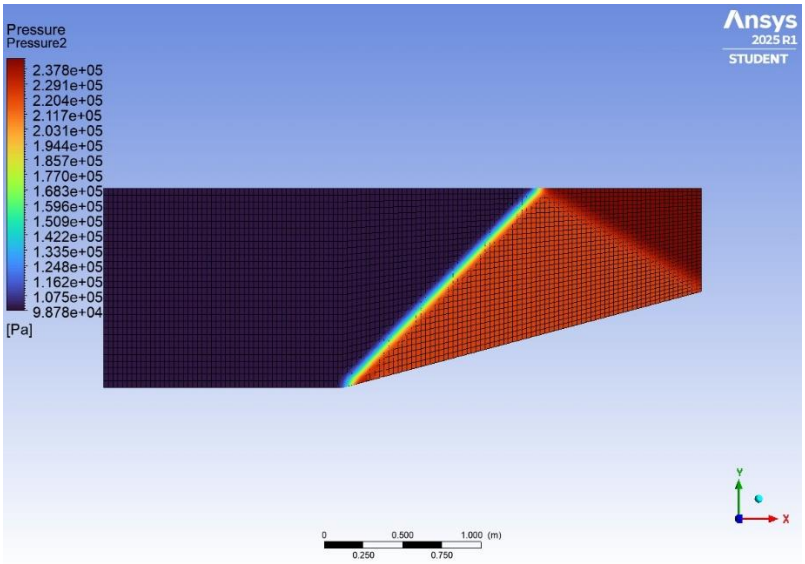

Figure 6: Pressure contour for medium mesh (0.03 m)

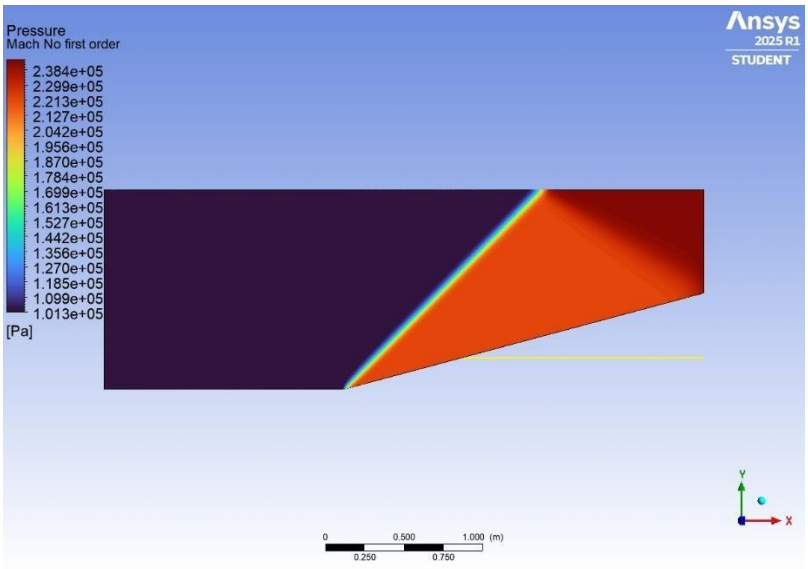

Figure 7: Pressure contour for finer mesh (0.01 m)

6.3 Line Plots

Pressure vs. x-location plots confirmed the shock location and strength. The fine mesh provided a sharper pressure jump, while the first-order solution slightly smeared the transition. Across all three mesh sizes, the location and magnitude of the pressure jump were consistent, indicating a robust prediction of the shock. However, differences emerged in the smoothness of the plots after the shock rise.

In the coarse mesh, significant oscillations appeared immediately after the sharp pressure rise, indicating numerical artifacts and lack of resolution. These oscillations are typical in coarse meshes due to insufficient cell density near steep gradients. In contrast, the fine mesh (0.01 m) displayed a clean pressure transition with minimal overshoot or undershoot after the shock, reflecting higher numerical stability and improved flow resolution.

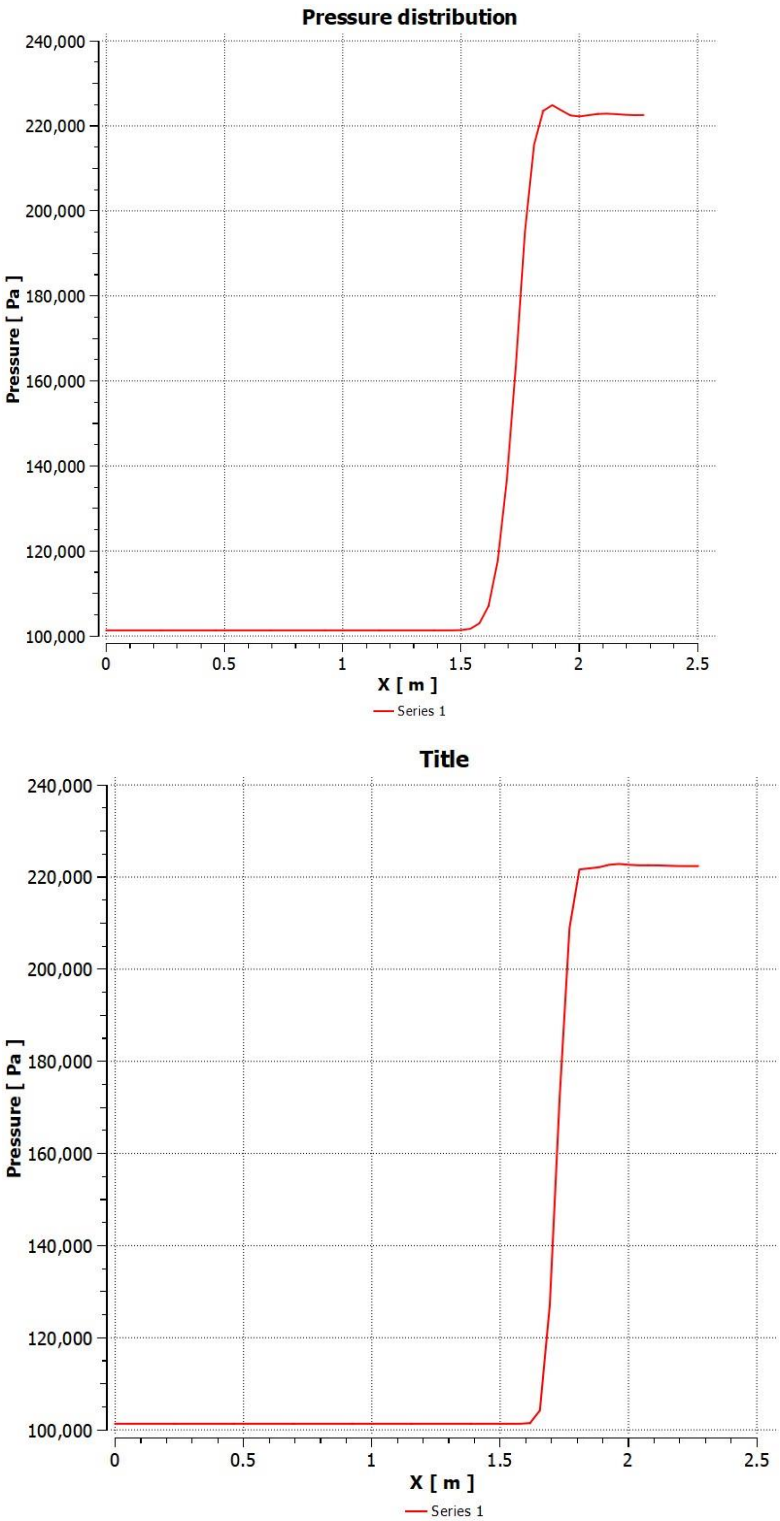

Figure 8: Static pressure distribution for coarse mesh (0.05 m) and medium mesh (0.03 m)

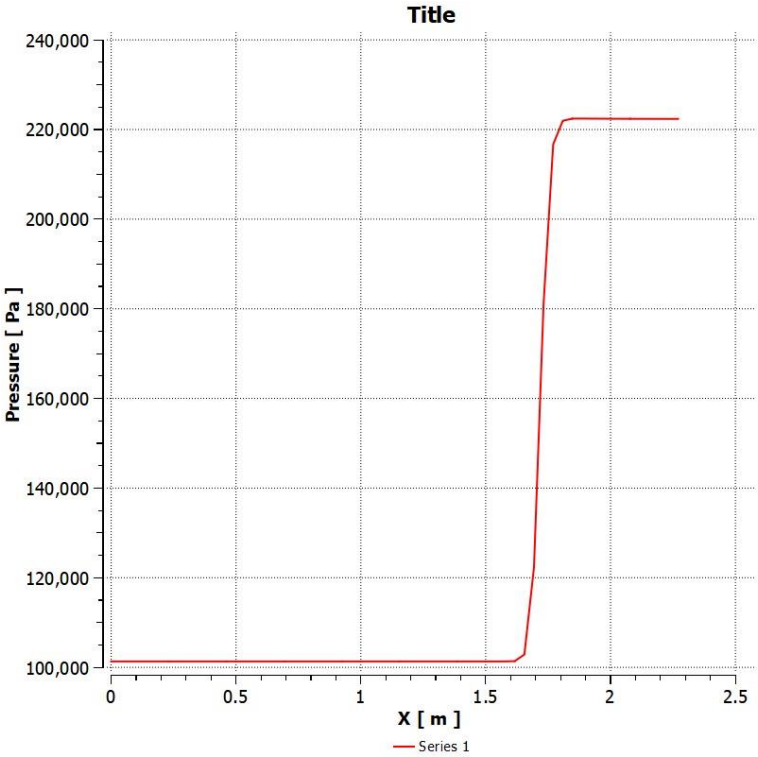

Figure 9: Static pressure distribution for finer mesh (0.01 m)

6.4 Comparison with Literature

Source	Shock Angle	Pressure Ratio	Notes
CFD (Medium)	~40.3°	~2.15	Sharp, stable shock
NASA (2024)	40.3°	~2.14	Euler, validated CFD
MDPI (2024)	40.3°	~2.1	Slight diffusion in solution

Agreement between CFD results and literature validates the simulation and supports the accuracy of ANSYS Fluent for modelling oblique shocks.

Conclusion

This project successfully modelled the supersonic flow over a 15° wedge using ANSYS Fluent and validated the results against published literature. Oblique shock angles and pressure distributions matched well with theoretical values and reference papers. A grid independence study confirmed that medium mesh resolution was sufficient for accuracy. Solver sensitivity was addressed by adjusting discretization and Courant number for

convergence. The work demonstrates the robustness of CFD tools in modelling fundamental compressible flow phenomena.

References

1. NASA. (2024). Supersonic Wedge Validation Case. Retrieved from: https://www.grc.nasa.gov/www/wind/valid/wedgeM2/wedgeM2.html
2. Mishra, A., et al. (2024). Numerical Investigation of Supersonic Flow Over a Wedge. *Mathematics*, 12(9), Article 1282. https://www.mdpi.com/2227-7390/12/9/1282
3. Ella, Hamza, (2025). AERO 4304 – Term Project W25. AERO 4304 - Term Project W25 - AERO4304A Computational Fluid Dynamics (LEC) Winter 2025